```
WASTE MANAGEMENT DATA CLEANING
```

```
import pandas as pd

df = pd.read_csv('/content/Waste_Dataset_V1.csv')
display(df.head())
```

	ItemName	Brand	Category	StockQty	ArrivalDate	ExpiryDate	DaysUntilExpiry	DailySaleAvg	SpoilageChance	SuggestedAction	StoreLocation	IsSpoiled	OnPromotion	Temperature!
0	Bingo Mad Angles	ITC	Snacks	92	20-06-2025	28-06-2025	0.5	12.1	1.0	donate	Chennai	False	No	
1	Multigrain Cookies	Sunfeast	Bakery	224	18-06-2025	29-06-2025	1.0	9.0	1.0	donate	Kanpur	True	No	
2	Nestlé Yogurt 150g	Nestlé	Dairy	201	23-06-2025	26-06-2025	0.5	20.1	1.0	donate	Mumbai	False	Yes	
3	Tomato 1kg	Local Farmer	Produce	204	18-06-2025	21-06-2025	0.5	9.7	1.0	donate	Kanpur	False	Yes	
4	Banana 6pcs	Local Farmer	Produce	154	14-06-2025	21-06-2025	0.5	5.6	1.0	donate	Mumbai	True	Yes	

```
df['ArrivalDate'] = pd.to_datetime(df['ArrivalDate'], format='%d-%m-%Y', errors='coerce')
df['ExpiryDate'] = pd.to_datetime(df['ExpiryDate'], format='%d-%m-%Y', errors='coerce')

print("Data types after conversion:")
print(df[['ArrivalDate', 'ExpiryDate']].info())
```

```
Data types after conversion:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 2 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   ArrivalDate  1000 non-null   datetime64[ns]
 1   ExpiryDate   1000 non-null   datetime64[ns]
dtypes: datetime64[ns](2)
memory usage: 15.8 KB
None
```

```
display(df.head())
```

	ItemName	Brand	Category	StockQty	ArrivalDate	ExpiryDate	DaysUntilExpiry	DailySaleAvg	SpoilageChance	SuggestedAction	StoreLocation	IsSpoiled	OnPromotion	Temperature!
0	Bingo Mad Angles	ITC	Snacks	92	2025-06-20	2025-06-28	0.5	12.1	1.0	donate	Chennai	False	No	
1	Multigrain Cookies	Sunfeast	Bakery	224	2025-06-18	2025-06-29	1.0	9.0	1.0	donate	Kanpur	True	No	
2	Nestlé Yogurt 150g	Nestlé	Dairy	201	2025-06-23	2025-06-26	0.5	20.1	1.0	donate	Mumbai	False	Yes	
3	Tomato 1kg	Local Farmer	Produce	204	2025-06-18	2025-06-21	0.5	9.7	1.0	donate	Kanpur	False	Yes	
4	Banana 6pcs	Local Farmer	Produce	154	2025-06-14	2025-06-21	0.5	5.6	1.0	donate	Mumbai	True	Yes	

```
df['OnPromotion'] = df['OnPromotion'].map({'Yes': True, 'No': False})

print("Data types after conversion:")
print(df[['OnPromotion']].info())
```

```
Data types after conversion:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 1 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   OnPromotion  1000 non-null   bool
dtypes: bool(1)
memory usage: 1.1 KB
None
```

```
display(df.head())
```

	ItemName	Brand	Category	StockQty	ArrivalDate	ExpiryDate	DaysUntilExpiry	DailySaleAvg	SpoilageChance	SuggestedAction	StoreLocation	IsSpoiled	OnPromotion	Temperature!
0	Bingo Mad Angles	ITC	Snacks	92	2025-06-20	2025-06-28	0.5	12.1	1.0	donate	Chennai	False	False	
1	Multigrain Cookies	Sunfeast	Bakery	224	2025-06-18	2025-06-29	1.0	9.0	1.0	donate	Kanpur	True	False	
2	Nestlé Yogurt 150g	Nestlé	Dairy	201	2025-06-23	2025-06-26	0.5	20.1	1.0	donate	Mumbai	False	True	
3	Tomato 1kg	Local Farmer	Produce	204	2025-06-18	2025-06-21	0.5	9.7	1.0	donate	Kanpur	False	True	
4	Banana 6pcs	Local Farmer	Produce	154	2025-06-14	2025-06-21	0.5	5.6	1.0	donate	Mumbai	True	True	

```
for col in ['Category', 'SuggestedAction', 'StoreLocation']:
    df[col] = df[col].astype('category')

print("Data types after conversion:")
print(df[['Category', 'SuggestedAction', 'StoreLocation']].info())
```

```
Data types after conversion:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 3 columns):
```

```
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Category         1000 non-null   category
 1   SuggestedAction  1000 non-null   category
 2   StoreLocation    1000 non-null   category
dtypes: category(3)
memory usage: 3.7 KB
None
```

```
display(df.head())
```

	ItemName	Brand	Category	StockQty	ArrivalDate	ExpiryDate	DaysUntilExpiry	DailySaleAvg	SpoilageChance	SuggestedAction	StoreLocation	IsSpoiled	OnPromotion	TemperatureS
0	Bingo Mad Angles	ITC	Snacks	92	2025-06-20	2025-06-28	0.5	12.1	1.0	donate	Chennai	False	False	
1	Multigrain Cookies	Sunfeast	Bakery	224	2025-06-18	2025-06-29	1.0	9.0	1.0	donate	Kanpur	True	False	
2	Nestlé Yogurt 150g	Nestlé	Dairy	201	2025-06-23	2025-06-26	0.5	20.1	1.0	donate	Mumbai	False	True	
3	Tomato 1kg	Local Farmer	Produce	204	2025-06-18	2025-06-21	0.5	9.7	1.0	donate	Kanpur	False	True	
4	Banana 6pcs	Local Farmer	Produce	154	2025-06-14	2025-06-21	0.5	5.6	1.0	donate	Mumbai	True	True	

```
# Fill missing 'Zone' values based on 'StoreLocation'
df.loc[(df['StoreLocation'].isin(['Bangalore', 'Chennai'])) & (df['Zone'].isna()), 'Zone'] = 'South'
df.loc[(df['StoreLocation'] == 'Kanpur') & (df['Zone'].isna()), 'Zone'] = 'North'

# Fill remaining missing 'Zone' values if any, with a generic placeholder or based on further logic if needed.
# For now, let's just make sure it's a category type after filling some values
df['Zone'] = df['Zone'].astype('category')

# Fill missing 'TemperatureSupport' values with 'Not Recorded'
df['TemperatureSupport'] = df['TemperatureSupport'].fillna('Not Recorded')

print("Data types and non-null counts after filling missing values:")
print(df[['Zone', 'TemperatureSupport']].info())
```

```
Data types and non-null counts after filling missing values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 2 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Zone                1000 non-null   category
 1   TemperatureSupport  1000 non-null   object
dtypes: category(1), object(1)
memory usage: 9.0+ KB
None
```

```
display(df.head())
```

	ItemName	Brand	Category	StockQty	ArrivalDate	ExpiryDate	DaysUntilExpiry	DailySaleAvg	SpoilageChance	SuggestedAction	StoreLocation	IsSpoiled	OnPromotion	TemperatureS
0	Bingo Mad Angles	ITC	Snacks	92	2025-06-20	2025-06-28	0.5	12.1	1.0	donate	Chennai	False	False	
1	Multigrain Cookies	Sunfeast	Bakery	224	2025-06-18	2025-06-29	1.0	9.0	1.0	donate	Kanpur	True	False	
2	Nestlé Yogurt 150g	Nestlé	Dairy	201	2025-06-23	2025-06-26	0.5	20.1	1.0	donate	Mumbai	False	True	
3	Tomato 1kg	Local Farmer	Produce	204	2025-06-18	2025-06-21	0.5	9.7	1.0	donate	Kanpur	False	True	
4	Banana 6pcs	Local Farmer	Produce	154	2025-06-14	2025-06-21	0.5	5.6	1.0	donate	Mumbai	True	True	

```
date_difference = df['ExpiryDate'] - df['ArrivalDate']

# Check if any ExpiryDate is not greater than ArrivalDate
invalid_dates_count = date_difference[date_difference <= pd.Timedelta(seconds=0)].count()

if invalid_dates_count == 0:
    print("Verification successful: ExpiryDate is always greater than ArrivalDate.")
else:
    print(f"Verification failed: Found {invalid_dates_count} instances where ExpiryDate is not greater than ArrivalDate.")
    print("These instances are:")
    display(df[date_difference <= pd.Timedelta(seconds=0)])
```

```
Verification successful: ExpiryDate is always greater than ArrivalDate.
```

```
df['DaysUntilExpiry'] = (df['ExpiryDate'] - df['ArrivalDate']).dt.days

print("Data types after recalculation:")
print(df[['ArrivalDate', 'ExpiryDate', 'DaysUntilExpiry']].info())
```

```
Data types after recalculation:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 3 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   ArrivalDate      1000 non-null   datetime64[ns]
 1   ExpiryDate       1000 non-null   datetime64[ns]
 2   DaysUntilExpiry  1000 non-null   int64
dtypes: datetime64[ns](2), int64(1)
memory usage: 23.6 KB
None
```

```
display(df.head())
```

	ItemName	Brand	Category	StockQty	ArrivalDate	ExpiryDate	DaysUntilExpiry	DailySaleAvg	SpoilageChance	SuggestedAction	StoreLocation	IsSpoiled	OnPromotion	Temperature
0	Bingo Mad Angles	ITC	Snacks	92	2025-06-20	2025-06-28	8	12.1	1.0	donate	Chennai	False	False	
1	Multigrain Cookies	Sunfeast	Bakery	224	2025-06-18	2025-06-29	11	9.0	1.0	donate	Kanpur	True	False	
2	Nestlé Yogurt 150g	Nestlé	Dairy	201	2025-06-23	2025-06-26	3	20.1	1.0	donate	Mumbai	False	True	
3	Tomato 1kg	Local Farmer	Produce	204	2025-06-18	2025-06-21	3	9.7	1.0	donate	Kanpur	False	True	
4	Banana 6pcs	Local Farmer	Produce	154	2025-06-14	2025-06-21	7	5.6	1.0	donate	Mumbai	True	True	

```
median_days_until_expiry = df['DaysUntilExpiry'].median()
pct_90_days_until_expiry = df['DaysUntilExpiry'].quantile(0.90)

print(f"Median DaysUntilExpiry: {median_days_until_expiry}")
print(f"90th Percentile DaysUntilExpiry: {pct_90_days_until_expiry}")

outliers_df = df[df['DaysUntilExpiry'] > pct_90_days_until_expiry]

print(f"\nNumber of outliers (DaysUntilExpiry > 90th percentile): {len(outliers_df)}")
print("\nOutliers by Category (DaysUntilExpiry > 90th percentile):")
display(outliers_df.groupby('Category')['DaysUntilExpiry'].agg(['count', 'min', 'max', 'mean']))

print("\nSample of outlier data:")
display(outliers_df.head())
```

```
Median DaysUntilExpiry: 7.0
90th Percentile DaysUntilExpiry: 11.0

Number of outliers (DaysUntilExpiry > 90th percentile): 93

Outliers by Category (DaysUntilExpiry > 90th percentile):
/tmp/ipython-input-194039701.py:11: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to ret
  display(outliers_df.groupby('Category')['DaysUntilExpiry'].agg(['count', 'min', 'max', 'mean']))
```

Category	count	min	max	mean
Bakery	16	12.0	12.0	12.0
Dairy	20	12.0	12.0	12.0
Meat	28	12.0	12.0	12.0
Produce	0	NaN	NaN	NaN
Snacks	29	12.0	12.0	12.0

```
Sample of outlier data:
```

	ItemName	Brand	Category	StockQty	ArrivalDate	ExpiryDate	DaysUntilExpiry	DailySaleAvg	SpoilageChance	SuggestedAction	StoreLocation	IsSpoiled	OnPromotion	Temperature
14	Brown Bun Pack	Harvest Gold	Bakery	90	2025-06-26	2025-07-08	12	22.4	0.08	none	Chennai	False	True	
26	Mutton Mince 250g	Godrej	Meat	96	2025-06-19	2025-07-01	12	22.0	0.29	none	Delhi	False	True	
27	Kurkure Masala	PepsiCo	Snacks	224	2025-06-24	2025-07-06	12	3.6	1.00	donate	Bangalore	True	True	
42	Mutton Mince 250g	Godrej	Meat	271	2025-06-22	2025-07-04	12	8.9	1.00	donate	Delhi	False	False	
43	Amul Milk 500ml	Amul	Dairy	11	2025-06-16	2025-06-28	12	14.1	0.31	none	Chennai	False	True	

```
display(df.head())
```

	ItemName	Brand	Category	StockQty	ArrivalDate	ExpiryDate	DaysUntilExpiry	DailySaleAvg	SpoilageChance	SuggestedAction	StoreLocation	IsSpoiled	OnPromotion	Temperature
0	Bingo mad angles	Itc	Snacks	92	2025-06-20	2025-06-28	8	12.1	1.0	donate	Chennai	False	False	
1	Multigrain cookies	Sunfeast	Bakery	224	2025-06-18	2025-06-29	11	9.0	1.0	donate	Kanpur	True	False	
2	Nestlé yogurt 150g	Nestlé	Dairy	201	2025-06-23	2025-06-26	3	20.1	1.0	donate	Mumbai	False	True	
3	Tomato 1kg	Local farmer	Produce	204	2025-06-18	2025-06-21	3	9.7	1.0	donate	Kanpur	False	True	
4	Banana 6pcs	Local farmer	Produce	154	2025-06-14	2025-06-21	7	5.6	1.0	donate	Mumbai	True	True	

```
df['ItemName'] = df['ItemName'].str.strip().str.capitalize()
df['Brand'] = df['Brand'].str.strip().str.capitalize()

print("Unique values for ItemName after cleaning (first 10):\n", df['ItemName'].unique()[:10])
print("\nUnique values for Brand after cleaning (first 10):\n", df['Brand'].unique()[:10])
```

```
Unique values for ItemName after cleaning (first 10):
 ['Bingo mad angles' 'Multigrain cookies' 'Nestlé yogurt 150g' 'Tomato 1kg'
 'Banana 6pcs' 'Lays chips 100g' 'Mutton mince 250g' 'Amul milk 500ml'
 'Chicken breast 500g' 'Brown bun pack']

Unique values for Brand after cleaning (first 10):
 ['Itc' 'Sunfeast' 'Nestlé' 'Local farmer' 'Pepsico' 'Godrej' 'Amul'
 'Harvest gold' 'Mother dairy' 'Freshcatch']
```

```
df.to_csv('Waste_Dataset_Processed.csv', index=False)
print("DataFrame exported to 'Waste_Dataset_Processed.csv' successfully.")
```

```
DataFrame exported to 'Waste_Dataset_Processed.csv' successfully.
```